

Requirements Document

1. Application Overview

Application Name: E-Commerce Storefront with Wishlist

Description: A web-based e-commerce storefront application that allows users to browse products, search and filter by category, and manage multiple wishlists. The application features a core wishlist merge capability with automatic duplicate detection. All data is persisted locally using browser LocalStorage.

2. Users and Usage Scenarios

Target Users: Online shoppers who want to organize and manage product wishlists for future purchase consideration.

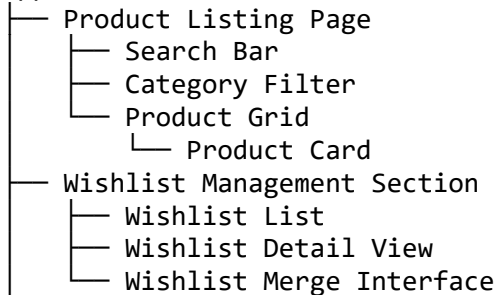
Core Usage Scenarios:

- Browse and discover products across different categories
 - Search for specific products by name or description
 - Create and manage multiple wishlists for different purposes (e.g., Birthday Ideas, Home Decor, Gifts)
 - Consolidate wishlists by merging two lists into one without duplicates
 - Access saved wishlists across browser sessions
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3. Page Structure and Functionality

Page Hierarchy

Application Root



3.1 Product Listing Page

Purpose: Display all available products with search and filter capabilities.

Components:

- **Search Bar:** Input field for searching products by name or description
- **Category Filter:** Dropdown or button group to filter products by category
- **Product Grid:** Responsive grid layout displaying product cards

Functionality:

- Display all products in grid format
- Search products: filter results based on user input in real-time
- Filter by category: show only products matching selected category

- Show loading state while data is being retrieved
- Show empty state when no products match search/filter criteria

3.2 Product Card

Display Information:

- Product image
- Product name
- Product price
- Product category

Actions:

- Add to Wishlist button: opens wishlist selection interface

3.3 Wishlist Management Section

Purpose: Create, view, edit, and manage wishlists.

Functionality:

- Create new wishlist: user provides wishlist name
- View all wishlists: display list of all created wishlists with name and item count
- Rename wishlist: edit wishlist name
- Delete wishlist: remove wishlist and all associated items
- View wishlist details: display all products in selected wishlist
- Remove product from wishlist: delete specific product from wishlist

3.4 Add Product to Wishlist Interface

Trigger: User clicks Add to Wishlist button on product card.

Functionality:

- Display list of existing wishlists
- Allow user to select target wishlist
- Option to create new wishlist if none exist or user wants a new one
- Add product to selected wishlist
- Show success message upon completion
- Show error message if product already exists in selected wishlist

3.5 Wishlist Merge Interface

Purpose: Merge two wishlists into one with automatic duplicate detection.

Functionality:

- Display all available wishlists
- User selects source wishlist (wishlist to merge from)
- User selects target wishlist (wishlist to merge into)
- Validation: source and target must be different wishlists
- Show merge preview:
 - Source wishlist name and product count
 - Target wishlist name and current product count
 - Estimated post-merge product count (after deduplication)
- Confirm merge action
- Execute merge:
 - Add all products from source to target (skip duplicates based on product_id)
 - Delete source wishlist
 - Update target wishlist timestamp
- Show merge result:
 - Display updated target wishlist
 - Show count of new items added
 - Show success message

- Show error message if merge fails
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4. Business Rules and Logic

4.1 Product Management

- Products are pre-defined and stored in LocalStorage
- Each product has unique id, name, description, price, image_url, category

4.2 Wishlist Management

- Users can create unlimited wishlists
- Each wishlist has unique id, name, created_at timestamp, updated_at timestamp
- Wishlist names must be non-empty
- Wishlist names can be duplicated (no uniqueness constraint)

4.3 Wishlist Items

- Each wishlist item links a product to a wishlist
- Unique constraint: one product can only appear once per wishlist (wishlist_id + product_id combination must be unique)
- Each item has id, wishlist_id, product_id, added_at timestamp
- Adding duplicate product to same wishlist shows error message

4.4 Wishlist Merge Rules

- User must select two different wishlists (source and target)
- Merge operation is atomic: either all changes succeed or none apply
- Deduplication logic: products are identified by product_id; if product already exists in target wishlist, skip adding from source
- After successful merge:
 - All unique products from source are added to target
 - Source wishlist is deleted
 - Target wishlist updated_at timestamp is updated
- Merge preview calculates estimated post-merge count: $\text{target_count} + (\text{source_count} - \text{overlap_count})$

4.5 Data Persistence

- All data (products, wishlists, wishlist items) stored in browser LocalStorage
- Data automatically loaded on application start
- Data persists across browser sessions
- Data is lost if LocalStorage is cleared

4.6 Search and Filter Logic

- Search: case-insensitive matching against product name and description
 - Filter: exact match on product category
 - Search and filter can be applied simultaneously (AND logic)
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5. Exception and Edge Cases

Scenario	Handling
No products available	Show empty state message: No products available
Search returns no results	Show empty state message: No products match your search
Filter returns no results	Show empty state message: No products in this category
User tries to create wishlist with empty name	Show error message: Wishlist name cannot be empty
User tries to add product already in wishlist	Show error message: Product already in this wishlist
User tries to merge wishlist with itself	Show error message: Cannot merge wishlist with itself
User tries to merge when only one wishlist exists	Disable merge action or show message: Need at least two wishlists to merge
User deletes last wishlist	Show empty state: No wishlists yet. Create one to get started
LocalStorage data is corrupted or missing	Initialize with empty state and show appropriate empty state messages
Merge operation fails mid-process	Rollback all changes, show error message: Merge failed. Please try again
User removes last product from wishlist	Wishlist remains but shows empty state: This wishlist is empty

6. Acceptance Criteria

1. User opens application and sees product listing page with all products displayed in grid layout
 2. User searches for a product by name, and matching products are displayed
 3. User filters products by category, and only products in that category are shown
 4. User clicks Add to Wishlist on a product, selects or creates a wishlist, and product is added successfully
 5. User navigates to wishlist management section and views all created wishlists
 6. User opens a wishlist and sees all products added to it
 7. User initiates wishlist merge, selects source and target wishlists, views merge preview showing product counts
 8. User confirms merge, and target wishlist now contains all unique products from both wishlists, source wishlist is deleted, and success message shows number of new items added
 9. User refreshes browser, and all wishlists and products persist correctly
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7. Out of Scope for This Release

- User authentication and login system
- Shopping cart functionality
- Checkout and payment processing
- Backend server or database integration
- Product inventory management
- Product reviews and ratings
- Wishlist sharing with other users
- Export wishlist to external formats

- Product recommendations
- Price tracking or alerts
- Multi-language support
- Dark mode theme
- Wishlist sorting or reordering
- Bulk operations (e.g., add multiple products at once)
- Undo/redo functionality
- Product comparison feature